MICROBURBS HACKATHON · TASK 4 · TEAM-OPAL

Would you let one flat vote ten times?

We price every City of Sydney flat from **its own past rent** and from **its neighbours in the same building** — one flat, one vote — and we only trust evidence that is still fresh.

$45
typical miss on 2,898 held-out adverts (Jun–Jul 2026)

77%
within $100 a week

0.88
R^2 on the same held-out adverts

65,777
flats priced, with and without a car space

Reference points published by the organisers (1,699 Aug–Sep adverts): simple estimate $100, with parking $93, Microburbs rental valuation $86 (55% within $100). Different test adverts — the official score comes from Aug–Sep.

Trained only on adverts dated ≤ cut-off; validated by date, never at random

THE PROBLEM

Same building. Same bedrooms. Up to $500 apart.

45 Shelley St, Sydney — one-bedroom flats, latest advert since 2025:

floor	water visible (share of directions)	asking rent
2–3	0%	$700 – $875
4–5	3–17%	$685 – $850
6–7	25–33%	$875 – $1,200

Water visibility measured per floor from NSW LiDAR line-of-sight (36 directions, 1.5 km).

Suburb averages can't see this. A postcode × bedroom rule from government bond data misses by **$100** a week.

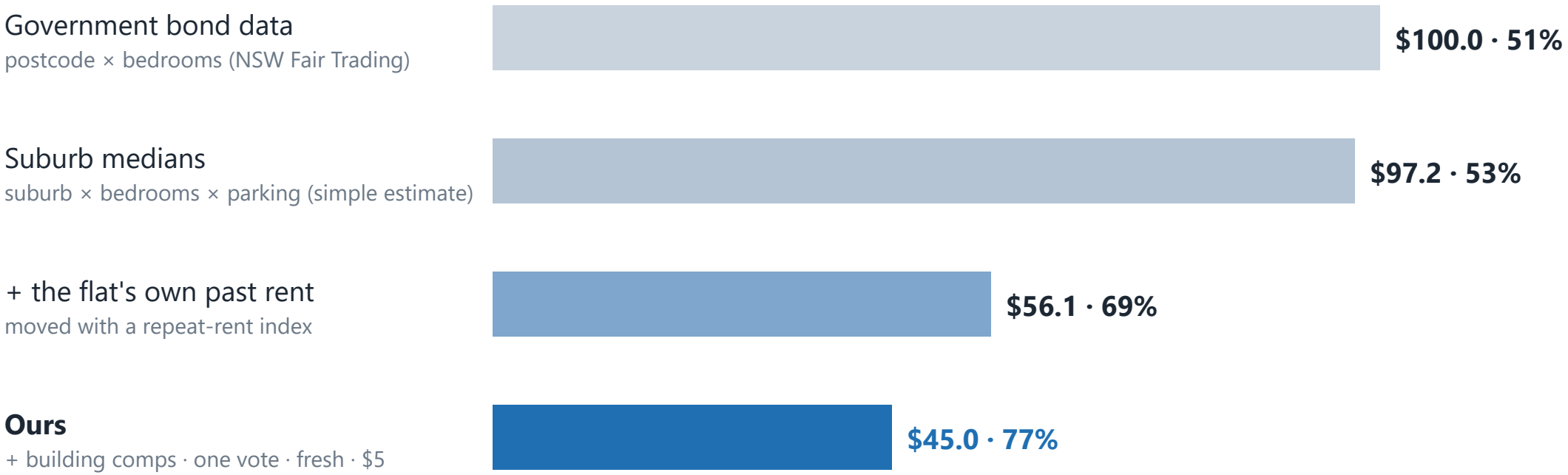
The flat itself can. Its own past rent and the rents of its neighbours already contain the floor, the view, the noise, the renovation.

The hard part is deciding **which evidence to trust**: stale adverts, relisted flats, changed bedroom counts.

Source: supplied past leases + Microburbs API adverts; LiDAR © NSW Spatial Services (CC BY)

HELD-OUT RESULT · WINDOW B (TRAIN ≤ 31 MAY, TEST JUN–JUL 2026, 2,898 ADVERTS)

From $100 to $45 — each step on the same adverts



bar = median absolute error; % = share within $100 · bond data restricted to files published before the cut-off

Window A (May) shows the same ladder: $100 → $92 → $57 → $45

METHOD

Four steps, all rebuilt from adverts before each cut-off

1. **Repeat-rent index.** Pairs of adverts of the *same* flat give the market move — location and quality cancel out. Held flat after July (Sydney unit asking rents stalled mid-2026).
2. **Own history.** The flat's latest advert *with the same bedroom count*, moved to today with the index. Adverts older than 3 years are dropped as stale.
3. **Same-building comparables, one flat one vote.** Other flats in the building with the same bedrooms; each neighbour counts once however often it relisted; the target is excluded; shrunk toward suburb medians when neighbours are few.
4. **Floor view and the grid.** LiDAR floor-view difference against those neighbours (coefficients estimated within buildings); parking premium for the second column; snapped to the $5 asking grid.

Rules chosen on window A (May), frozen, then evaluated on window B (Jun–Jul); final model trained on everything ≤ 31 Jul

WHAT WE FOUND

Three non-obvious things the data told us

Old ≠ useful

A flat's own advert from more than 3 years ago is **worse than its neighbours' current rents**: $53 → $46.5 on window A when we swapped them. 75% of repeat flats last advertised over a year ago.

Same address ≠ same flat

When the bedroom count changes between adverts (whole-building ids, split lettings) the error **multiplies by ~2.4**. Own history only counts when bedrooms match.

Rents live on a grid

84% of asking rents end in $0, 52% are multiples of $50, 26% of $100. Snapping to $5 took window A from **$46.5 to $45**.

And one flat that relisted ten times no longer outvotes nine neighbours that advertised once.

All numbers measured on real adverts with the date split

PUBLIC DATA, HONESTLY

Does the market pay for the view? Yes — inside the building.

NSW LiDAR 2013 + 2020 → height of 4,399 buildings, 79 built between surveys, and a line-of-sight test for every floor: water visible from 6% of floors 1–2, 60% of floors above 20.

within the same building × bedrooms	rent
water in every direction vs none	+16 to +19%
park in every direction vs none	+4 to +6%
clear of every roof within 100 m	+5%

22,593 flats in 2,901 building × bedroom groups; re-estimated in each window.

What did not work: building-level layers (water +25%, age −6% per decade, flight path −2%, new build −4% against suburb medians) **made errors worse** — the neighbours' rents already contain them. Rejected on window A.

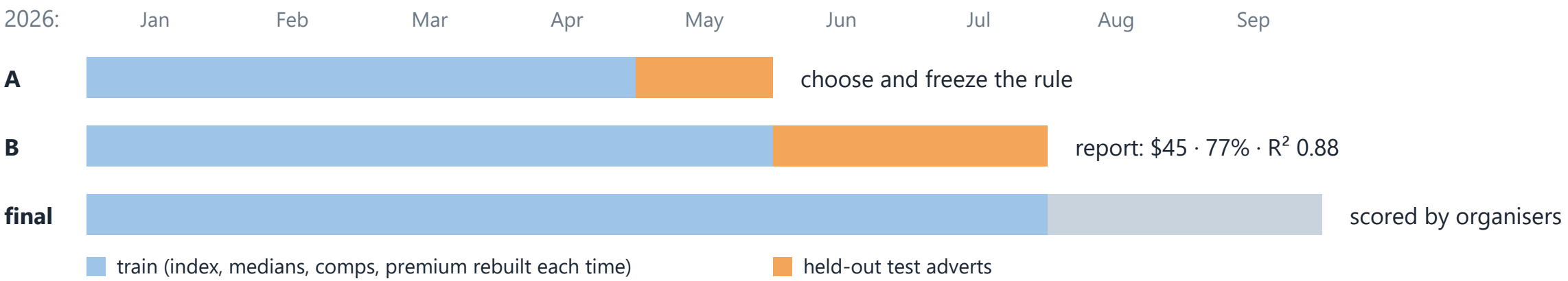
What we kept: only the floor-view *difference* to the comparable flats. Window A: share within $100 79.1% → 79.2%; window B neutral ($45, 77.1%), R^2 0.877 → 0.880.

Heat, canopy, flight path: insight maps, residuals within ±2% after the model.

LiDAR © NSW Spatial Services (CC BY) · OSM © contributors (ODbL)

VALIDATION

Built only on earlier adverts, tested on later ones



- No random splits; the code refuses any training row dated after the window's cut-off.
- Candidates (one-vote strength, stale threshold, snapping, public layers) compared on A only; B evaluated with the frozen rule. The floor-view step was added on A after the base rule froze — both B results are reported.
- Our estimates sit about 1% above later adverts — consistent with Sydney unit asking rents flattening after July.

Cut-off: nothing dated after 31 July 2026 from any source

A rent you can check flat by flat

- **Tenants** see whether an asking rent is high for *this* floor of *this* building.
- **Landlords** set a rent that fits the flats around them — not the suburb average.
- **Investors** compare likely rent before they buy.

Next: unit orientation (which way the flat faces), strata sale prices within the building, and a live "evidence card" per estimate: own advert, neighbours used, floor-view adjustment.

Data and credits

Microburbs hackathon files and API (used on the day, not republished) · NSW Spatial Services LiDAR, cadastre, boundaries (CC BY) · OpenStreetMap (ODbL) · NSW DPHI heat and canopy (CC BY 4.0) · NSW DCCEEW urban heat island (CC BY) · ABS Census 2021 (CC BY 4.0) · NSW Fair Trading rental bond lodgements (baseline only)

Code, report.html and rents_Team-Opal.csv: Team-Opal repository.